omaly Detection -- Server Response Tim

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find

Domain: Time Series Analysis

Universe

- Zscore Detection: \{n_detected, indices, precision, recall, f1\}
- Iqr Detection: \{n_detected, indices, precision, recall, f1\}
- Combined Detection: \{n_detected, indices, precision, recall, f1\}
- Change Point: \{detected, best_match, true, error\}
- Regime Analysis: \{pre_change_mean, post_change_mean, t_statistic, p_value\}

Variables

Symbol	Meaning	Type / Range
x	decision variable	\mathbb{{R}}

Structure

- **Data**: 1,000 hourly response time observations (synthetic)
- **Regime 1** (obs 0--599): baseline mean 200ms, std 30ms, slight daily seasonality
- **Regime 2** (obs 600--999): elevated mean 350ms, std 30ms (infrastructure degradation)
- **Anomalies**: 15 injected spike anomalies at random positions (5x normal magnitude)
- **Questions**: (1) Which observations are anomalous spikes? (2) Where does the regime change occur?

Real-World Mapping

Real-World Concept	Mathematical Object
problem input	instance parameters

Constraints

(1) Anomaly detection**: identifying observations that deviate significantly from expected beh

(2) Change point detection**: finding the time index where the statistical properties of a series

(3) Z-score**: number of standard deviations from the (rolling) mean; threshold-based flagging

(4) IQR (interquartile range)**: nonparametric spread measure; outliers fall beyond 1.5*IQR fe

(5) PELT**: Pruned Exact Linear Time algorithm for optimal segmentation (Killick et al. 2012)

(6) Rolling statistics**: windowed mean/std that adapt to local behavior, preventing regime sh

(7) Precision / recall / F1**: standard classification metrics applied to anomaly detection perfor

Approach

Suggested approach: Z-score (rolling) + IQR + PELT (L2, BIC penalty)

Complexity class:

Available tools: Z-score

2. Solution

Answer:	Solution computed
Optimal:	No / Unknown
Feasible:	No
Algorithm:	Z-score (rolling) + IQR + PELT (L2, BIC penalty)
Time:	3.1975s

Solution Details

Method: 1. Z-score (rolling window): Compute rolling mean and std (window=50). Flag observations where |z-score| > 3.0. Adapts to local trends and is robust to gradual shifts.

2. IQR method (rolling window): For each observation, compute Q1, Q3, IQR over a local window (100 points). Flag observations beyond Q1 - 1.5*IQR or Q3 + 1.5*IQR fences. The rolling approach adapts to regime changes, unlike a global IQR. Simple, nonparametric, resistant to distributional assumptions.

3. PELT change point detection (ruptures library): Pruned Exact Linear Time algorithm with L2 (least squares) cost model and BIC penalty. Finds the optimal segmentation of the time series into regimes with different means.

Verification

✓	Recall Gte 0.7	Passed
✓	Precision Gte 0.5	Passed
✓	Change Point Within 20	Passed
✓	Few Change Points	Passed
✓	Regime Diff Significant	Passed
✓	Zscore Anomalies Extreme	Passed
✓	All Passed	Passed

3. Interpretation

The Question

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The Answer

A feasible solution was found.

What This Means

- Z-score anomaly detection: detects most injected spikes (recall >= 70%, precision >= 50%)
- IQR anomaly detection: complementary method with similar recall
- PELT change point: detected within 20 observations of the true change at obs 600
- Pre/post mean difference: statistically significant (t-test $p < 0.001$)
- All 6 verification checks pass

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Recommendations

1. Review the model assumptions to ensure real-world fidelity.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.
- Solutio